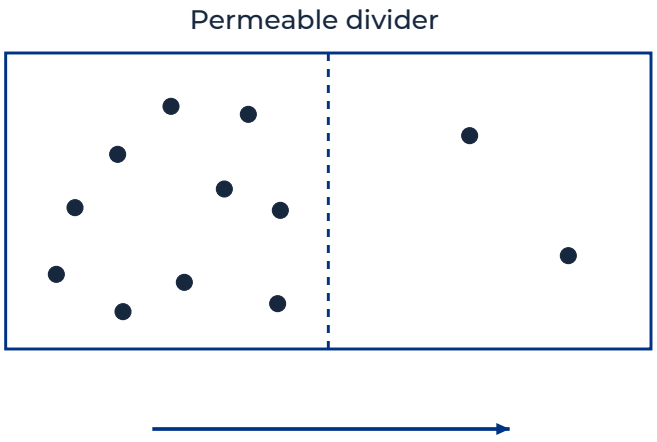
Diffusion and dynamic equilibrium

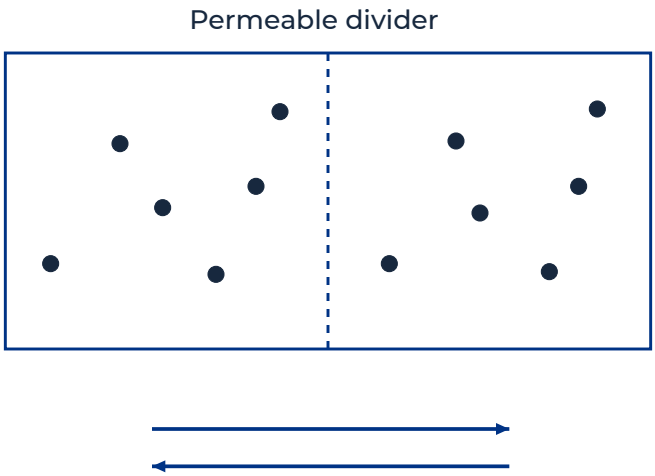
Random motion continues even when opposing average fluxes balance.

Initial gradient



Net: high -> low

Dynamic equilibrium



Equal opposing average fluxes

Idea	Meaning in this ideal model
Individual motion	Particles move randomly in both directions, including against the overall concentration gradient.
Net flux	Difference between directional transfers. Initially more crossings originate from the more concentrated side.
Equilibrium	No sustained net transfer; movement continues. Equal concentration is not always equal particle number.

Concentration = particle number / volume

Equal volumes + 12 total particles: average 6 on each side. Left volume = 2 x right volume: average 8 left and 4 right gives equal concentration. Counts fluctuate in a small system.

Assumptions: closed system, identical freely diffusing particles, fixed volumes, no reaction or electrical/pressure driving difference.

Diffusion and dynamic equilibrium

Self-check questions and explanations. Refer to the visual guide on the front.

With 10 particles left and 2 right in equal volumes, what is the initial net direction?

Initial net movement is left to right because 10 particles occupy the same volume as 2 on the other side. Random motion produces more crossings from the more concentrated side on average.

Can particles cross both ways while net flow has one direction?

Individual particles can cross either way. Net movement is the difference between the two directional transfers over a time interval, not a rule that each particle follows one direction.

What average counts are expected at equilibrium in equal volumes?

Average 6 left and 6 right. Small systems fluctuate, so an instantaneous count need not be exactly 6 and 6. Equal expected concentration is the equilibrium condition here.

If left volume is twice right volume, how do 12 particles distribute on average?

Average 8 left and 4 right. If right volume is V, left is 2V: 8/(2V) = 4/V. Equal counts would produce different concentrations.

Do particles stop at equilibrium?

Particles continue moving and crossing. At dynamic equilibrium the opposing average transfer rates balance, so there is no sustained net transfer despite continued molecular motion.

How does raising temperature affect this ideal diffusion model?

Higher temperature generally increases molecular motion and diffusion rate in this model. Equal concentration remains the condition for this ideal freely diffusing solute. Assume fixed volumes, identical environment and no reaction or binding.